

DATE & TIME FUNCTIONS

Date Value

Every date in Excel has an associated **date value**, which is how Excel calculates the passage of time (using midnight on 1/1/1900 as the starting point)

Excel recognizes most typed dates and automatically applies a common format (i.e. m/d/yyyy), along with an associated date value (cell format → General)

***Note:** If you type a date in a format that Excel does NOT recognize, it will be treated as text and there will be no associated date value; however, you can use a **DATEVALUE** or **TIMEVALUE** function to convert unformatted dates or times into serial values*

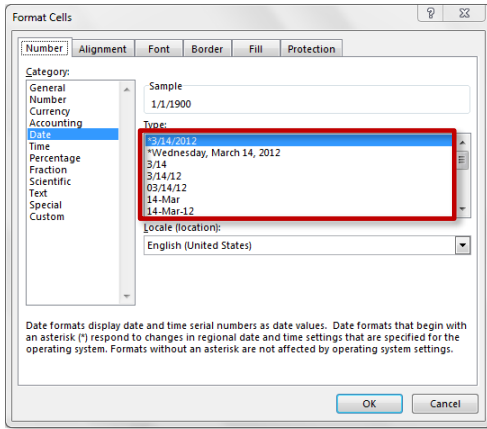
Date	Date Value
1/1/1900	1
1/11/1900	11
2/6/2015	42041
2/6/15 12:00 PM	42041.5
2/6/15 6:00 PM	42041.75

Jan 1, 1900 is the first date with an assigned date value (1). Feb 6, 2015 is the 42,041st day since 1/1/1900, so its date value = 42041

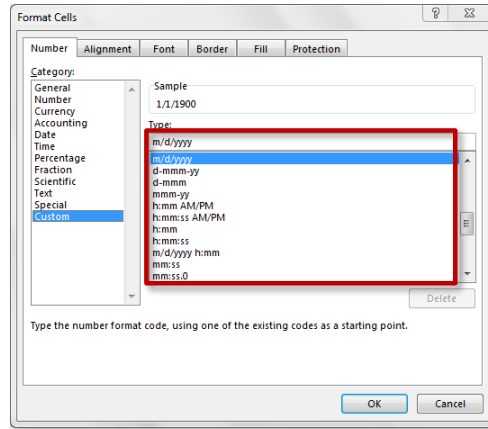
Date values can also indicate fractions of days: 42041.5 translates to noon on 2/6/2015 (50% through the day), and 42041.75 translates to 6:00pm on 2/6/2015 (75% through the day)

To format dates in Excel, you can either select a preset option from the “Date” category of the “Format Cells” dialog box, OR create your own **custom format**

Preset Formats:



Custom Format:

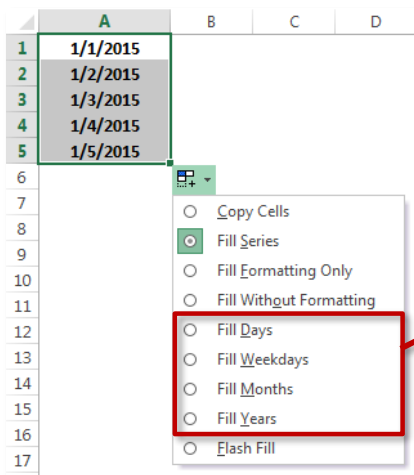


You can build your own custom formats using combinations of date/time codes. For example:

d = day w/out leading zero (1-31)
 dd = day w/ leading zero (01-31)
 ddd = day-of-week (Sat)
 dddd = day-of-week (Saturday)
 m = month w/out leading zero (1-15)
 mm = month w/ leading zero (01-15)
 mmm = month abbreviation (Jan)
 mmmm = full month (January)
 yy = last 2 digits of year (15)
 yyyy = full year (2015)

(full list available at support.office.com)

When you drag the corner of a cell containing a date, Excel automatically applies subsequent values automatically using **Fill Series** options:



Click the **Auto Fill Options** button to determine exactly which values your subsequent cells should take:

Copy Cells = Repeats the same value in all cells

Fill Days = Increases the date by 1 day per cell

Fill Weekdays = Increases the date by 1 day per cell (excluding weekends)

Fill Months = Increases the date by 1 month per cell

Fill Years = Increases the date by 1 year per cell

The **TODAY()** and **NOW()** functions return the current date or exact time

*Note: These are **volatile** functions, meaning that they change with every worksheet calculation*

TODAY()=	2/6/2015
NOW()=	2/6/2015 17:15

This is what the **TODAY()** and **NOW()** functions return at 5:15pm on February 6, 2015. Note that these values will automatically update with every change made to the workbook



PRO TIP:

Make sure to enter **TODAY()** and **NOW()** functions with both parentheses included – these functions don't refer to other cells

Serialization Formulas

Excel will always calculate dates and times based on their *precise* underlying serial values, but what if you need to work with less-specific values, like months instead of days, or hours instead of seconds?

The **YEAR**, **MONTH**, **DAY**, **HOUR**, **MINUTE**, and **SECOND** functions extract individual components of a given date:

	A	B	C	D	E	F	G
1		YEAR	MONTH	DAY	HOUR	MINUTE	SECOND
2	2/6/2015 17:57	2015	2	6	17	57	16
3		=YEAR(A2)	=MONTH(A2)	=DAY(A2)	=HOUR(A2)	=MINUTE(A2)	=SECOND(A2)
4							

Use the **EOMONTH** function to calculate the last day of a given month, or to calculate the start/end dates of previous or future months

=EOMONTH(start_date, months)

Reference to the cell containing the start/current date

Number of months before or after the start/current date (positive number yields a date in the future, negative number yields a date in the past)

	A	B	C
1			
2		Current Date:	8/3/2015
3			
4		End of month:	8/31/2015
5		Start of Month:	8/1/2015
6		Start of Next Month:	9/1/2015

=EOMONTH(C2, 0)
 =EOMONTH(C2, -1)+1
 =EOMONTH(C2, 0)+1

YEARFRAC calculates the fraction of a year represented by the number of whole days between two dates

=YEARFRAC(start_date, end_date, [basis])

Reference to the cell containing the start date

Reference to the cell containing the end date

Option specify the type of day count to use:

- 0 (default)** = US (NASD) 30/360
- 1** = actual/actual (**RECOMMENDED**)
- 2** = actual/360
- 3** = actual/365
- 4** = European 30/360

	A	B
1		
2	Start Date:	1/1/2015
3	End Date:	2/28/2015

=YEARFRAC(B2, B3, 1) = **15.9%**
 =YEARFRAC(B2, B3, 2) = **16.1%**



PRO TIP:

YEARFRAC is a great tool for pacing and projection calculations

If you want to know which day of the week a given date falls on, there are two ways to do it:

- 1) Use a custom cell format of either "ddd" (Sat) or "dddd" (Saturday)
*-Note that this doesn't change the underlying **value**, only how that value is displayed*
- 2) Use the **WEEKDAY** function to return a serial value corresponding to a particular day of the week (either 1-7 or 0-6)

=WEEKDAY(serial_number, [return type])

This refers to a cell containing a **date** or **time**

0 (default) = Sunday (1) to Saturday (7)

1 = Monday (1) to Sunday (7)

3 = Monday (0) to Sunday (6)

WORKDAY/NETWORKDAYS

WORKDAY returns a date that is a specified number of days before or after a given start date, excluding weekends and (optionally) holidays; **NETWORKDAYS** counts the number of workdays between two dates:

=WORKDAY(start_date, days, [holidays])

This refers to the cell containing the start date

Number of days before or after start date

Optional reference to a list of holiday dates

=NETWORKDAYS(start_date, end_date, [holidays])

This refers to the cell containing the start date

This refers to the cell containing the end date

Optional reference to a list of holiday dates

	A	B
1		
2	Start Date:	1/1/2015
3	End Date:	2/28/2015

=WORKDAY(B2, 20) = 1/29/2015

=NETWORKDAYS(B2, B3) = 42

DATEDIF calculates the number of days, months, or years between two dates

=DATEDIF(start_date, end_date, unit)

Reference to the cell
containing the start date

Reference to the cell
containing the end
date

How do you want to calculate the difference?

"D" = # of days between dates

"M" = # of months between dates

"Y" = # of years between dates

"MD" = # of days between dates, ignoring months and years

"YD" = # of days between dates, ignoring years

"YM" = # of months between dates, ignoring days and years

	A	B
1		
2	Start Date:	1/1/2015
3	End Date:	2/28/2015

=DATEDIF(B2, B3, "D") = 58

=DATEDIF(B2, B3, "MD") = 27



PRO TIP:

If you only need to calculate the # of days between dates, just use subtraction